

KCPC Week 3 Reading

The Lambda Calculus: Functions, Computation, and Numbers

Follow this guide alongside the slides and problem sheet. Start with the supplied Oxford notes, then use the accessible introductions for extra worked examples; the research bibliography in *Week 3 Papers* gives the original sources. Page numbers below refer to the *printed* pages of each text, rather than the PDF viewer counter.

I. The supplied further reading: Oxford lecture notes

Lambda Calculus and Types (Andrew D. Ker)

Oxford University Computing Laboratory, 16 lectures, Hilary Term 2009. Available locally at the supplied Oxford lecture-notes PDF (in `reading/`).

- **Chapter 1, §§1.2–1.5, pp. 2–7:** the grammar of terms, application association, free/bound variables, legal alpha-conversion and capture-avoiding substitution. Do this before Problems 1–5.
- **Chapter 4, §4.1, pp. 43–44:** Boolean selectors, Church numerals, successor, zero test and predecessor. Compare with Problems 6–10. The chapter’s introduction explains why numbers are encoded instead of built in.
- **Chapter 2, §§2.1–2.5, pp. 15–23 (after the workshop):** reduction sequences and the Church–Rosser theorem; why choosing different beta redexes need not change a normal form.
- **Chapter 3, §3.1, pp. 31–33 (extension):** reduction strategies and why a concrete evaluator must choose which redex to contract.

The appendix on computational practice (pp. 109–115) gives answers to the notes’ short calculation exercises; use it to check your attempts rather than to replace a proof.

II. Accessible introductions and worked examples

Lecture Notes on the Lambda Calculus (Peter Selinger)

Dalhousie University, revised 2013, arXiv:0804.3434. <https://arxiv.org/abs/0804.3434>

- **§2:** untyped terms, binding, substitution, alpha- and beta-conversion (Problems 1–5).
- **§3:** Church encodings of Booleans, numerals and arithmetic (Problems 6–10).
- **§4:** further theory of reduction; optional once the worksheet is complete.

Types and Programming Languages (Benjamin C. Pierce)

MIT Press, 2002. Contents and chapter locations: <https://www.cis.upenn.edu/~bcpierce/tapl/contents.pdf>

- **Chapter 5, §5.1, pp. 51–58:** untyped lambda syntax, substitution and evaluation; clarifies exactly what a binder does.
- **Chapter 5, §5.2, pp. 58–68:** currying, Booleans, pairs, Church numerals and recursion; alternate encoding examples.

- **Chapters 6–7, pp. 75–88 (optional):** nameless variables and an implementation of the calculus, particularly helpful if you want to build an evaluator after the two extended C++ homework tasks.

Church Numerals and Booleans (OpenDSA)

Virginia Tech, Programming Languages, §3.8. <https://opensa-server.cs.vt.edu/ODSA/Books/PL/html/ChurchNumerals.html> Interactive reduction traces for T/F, AND, successor, multiplication, predecessor and ISZERO. Use the animation to check a small input, then use iteration and induction to justify your answer for *every* numeral.

III. Suggested order

Read Ker §§1.2–1.5, solve Problems 1–5; read Ker §4.1 and OpenDSA §3.8, solve Problems 6–10; consult Pierce §5.2 for pair encodings, then prove the stack invariants in Problems 11–12. For original research and precise bibliographic locations see *Week 3 Papers*.

